indigestible material is stored in the large intestine till it is eliminated via the anus.

- Salivary glands, gastric glands, liver, pancreas, glands of Lieberkuhn and Brunner's glands contribute to the secretions of the GI tract.
- The mechanical processes involved in digestion are ingestion, mastication, deglutition, peristalsis, mixing and propulsion and defecation.

Digestion and absorption of carbohydrates

- Digestion of carbohydrates takes place briefly in the mouth and largely in the intestine. Amylases present in saliva and pancreatic juice and disaccharidases present in the brush border membrane of the intestine such as sucrase, lactase, maltase and isomaltase are responsible for digestion of carbohydrates.
- Monosaccharides are absorbed by both passive diffusion as well as active transport. Glucose is absorbed passively by a transporter, GLUT2 (Glucose Transporter 2). Active transport is mediated by a protein called as SGLT1 (Sodium Glucose transporter 1) in the luminal side and GLUT2 on the baso-lateral side.

Digestion and absorption of proteins

- In the stomach, HCl denatures the proteins and exposes the sites for action of the peptidases.
- Pepsin, Rennin, Gelatinase and Gastricsin in the stomach, enteropeptidase, trypsin, chymotrypsin, carboxypeptidases, elastases and gelatinases along with amino peptidases and carboxy peptidases contribute to the digestion of proteins.
- D-amino acids produced by bacteria are absorbed passively, while L amino acids are absorbed actively. Few di and tripeptides are also absorbed.
- Gamma glutamyl cycle is involved in the absorption of neutral amino acids.

Digestion and absorption of fats

- Fat digestion is facilitated by the detergent action of the Bile salts and micelle formation. Pancreatic lipase, cholesterol esterase and phospholipase A2 are the enzymes that complete lipid digestion. Fat is responsible for the satiety value of foods.
- Fats are absorbed by passive diffusion mainly in the proximal jejunum.

Digestion and absorption of nucleic acids

- No digestion of nucleic acids has been reported in the stomach.
- The nucleic acids are hydrolyzed to a mixture of nucleotides by ribonucleases and deoxy ribonucleases (both endo and exonucleases) present in pancreatic and intestinal secretions.

Gastrointestinal hormones

- Gastrin, Cholecystokinin, enterogastrone, secretin and motilin are the five major hormones that help in digestion.

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